

Collapsible Protective Case for Unmanned Aerial Vehicles

ABSTRACT

A protective case for an unmanned aerial vehicle includes a rigid base tray shaped to receive the vehicle's fuselage, a plurality of foam-lined wall panels hinged to the base tray, and a locking frame that holds the wall panels in an upright, closed configuration. Releasing the locking frame allows the wall panels to fold flat against the base tray for compact storage when the case is not in use.

BACKGROUND

Rigid cases for unmanned aerial vehicles provide good protection during transport but occupy substantial storage volume when empty, which is inconvenient for users who transport the vehicle frequently but store the case for extended periods between uses.

SUMMARY

In one embodiment, the base tray includes a contoured recess matching the underside of the vehicle fuselage, and four wall panels are each connected to an edge of the base tray by a piano hinge. A removable locking frame fits over the top edges of the upright wall panels to hold them in place during transport; removing the frame allows each panel to be folded down flat against the base tray, reducing the case's stored height substantially.

CLAIMS

1. A protective case comprising: a base tray having a recess shaped to receive a fuselage of an unmanned aerial vehicle; a plurality of wall panels each hingedly coupled to an edge of the base tray; and a removable locking frame configured to hold the plurality of wall panels in an upright configuration. 2. The case of claim 1, wherein each of the plurality of wall panels is coupled to the base tray by a piano hinge. 3. The case of claim 1, wherein removal of the locking frame permits each wall panel to fold flat against the base tray.